

Solutions: Vectors that fool intuition

Instructor copy · Vector addition and subtraction

PROBLEM 1

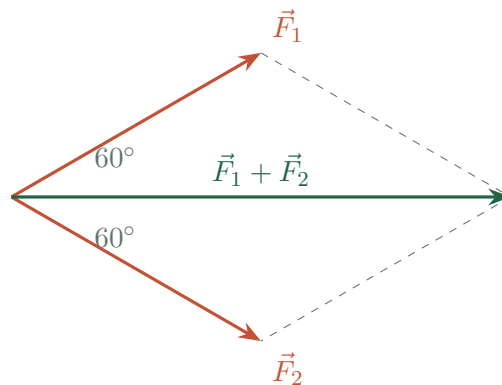
How can $600 + 600 = 600$?

Let each force make an angle θ with east. Symmetry makes the vertical components cancel. The horizontal components add:

$$600 \cos \theta + 600 \cos \theta = 600.$$

Therefore $2 \cos \theta = 1$, so $\theta = 60^\circ$. The two ropes lie 60° above and below east, giving an angle of

120° between the ropes.



For two 600 N pulls, the resultant can range from 0 to 1200 N. It is zero when the forces are opposite and 1200 N when they point together.

Answer. Vector addition combines components and directions. Only parallel vectors pointing the same way allow their magnitudes to be added directly.

Reasoning checkpoint. Ask students why an obtuse angle is already plausible before calculation: substantial sideways cancellation is needed for the sum to be no larger than one pull.

PROBLEM 2

Same sum. Same journey?

Both finish at $(3, 4)$ m after 7 s. Hence both have

$$\Delta\vec{r} = (3, 4) \text{ m}, \quad |\Delta\vec{r}| = 5 \text{ m}, \quad \vec{v}_{\text{avg}} = \left(\frac{3}{7}, \frac{4}{7}\right) \text{ m/s}.$$

Each walks 7 m, so each average speed is 1 m/s. The magnitude of each average velocity is only $5/7$ m/s.

For Aruzhan,

$$\vec{r}_A(t) = \begin{cases} (t, 0), & 0 \leq t \leq 3, \\ (3, t - 3), & 3 \leq t \leq 7. \end{cases}$$

For Daniyar,

$$\vec{r}_D(t) = \begin{cases} (0, t), & 0 \leq t \leq 4, \\ (t - 4, 4), & 4 \leq t \leq 7. \end{cases}$$

These positions are equal only at $t = 0$ and $t = 7$ s. Apart from the common start, they meet only at the finish.

At $t = 3.5$ s,

$$\vec{r}_A = (3, 0.5) \text{ m}, \quad \vec{r}_D = (0, 3.5) \text{ m},$$

so the vector from Aruzhan to Daniyar is

$$\vec{r}_D - \vec{r}_A = \boxed{(-3, 3) \text{ m}}.$$

Answer. $\vec{a} + \vec{b} = \vec{b} + \vec{a}$ guarantees the same resultant displacement and endpoint. It does not guarantee the same path or the same position at intermediate times.

PROBLEM 3

Two apps disagree—or do they?

App E uses the basis vectors east and north:

$$\vec{d} = 3 \hat{e}_{\text{east}} + 4 \hat{e}_{\text{north}}.$$

App N calls north its first positive direction and west its second. Therefore

$$\vec{d} = 4 \hat{e}_{\text{north}} - 3 \hat{e}_{\text{west}}.$$

The minus sign means 3 km opposite west, which is east. Converting App N's report to east–north coordinates gives

$$(4, -3)_{(\text{north}, \text{west})} = (3, 4)_{(\text{east}, \text{north})}.$$

The calculation $(3, 4) + (4, -3)$ combines coordinate entries attached to different basis directions. The first entries do not even refer to the same physical axis: one is east and the other is north.

Because the two reports describe the same physical displacement, their difference must be zero. After conversion,

$$(3, 4)_E - (3, 4)_E = \boxed{(0, 0)}.$$

Answer. Before adding or subtracting component pairs, express every vector in the same coordinate system, with the same axes, positive directions, and units.

Reasoning checkpoint. Changing axes changes the component numbers. It does not rotate the actual displacement or create a second displacement.

PROBLEM 4

Equal speeds on a collision course

B's position relative to A is

$$\vec{r}_{B/A}(0) = \vec{r}_B - \vec{r}_A = (8, -8) \text{ m.}$$

B's velocity relative to A is

$$\vec{v}_{B/A} = \vec{v}_B - \vec{v}_A = (0, 4) - (4, 0) = (-4, 4) \text{ m/s.}$$

Thus

$$\vec{r}_{B/A}(t) = (8, -8) + t(-4, 4).$$

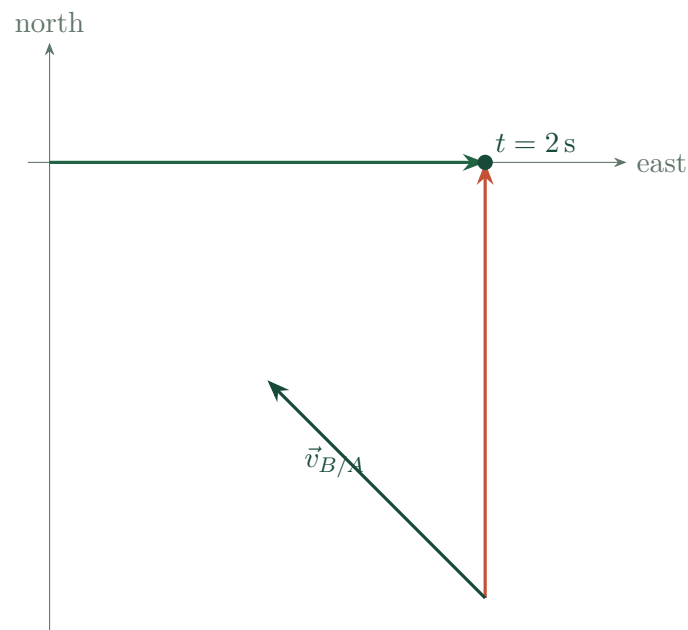
For a collision, both components must be zero at the same time:

$$8 - 4t = 0, \quad -8 + 4t = 0.$$

Both give $t = 2 \text{ s}$. The meeting position is

$$\vec{r}_A(2) = (0, 0) + (4, 0)(2) = \boxed{(8, 0) \text{ m}}.$$

Checking with B gives $(8, -8) + (0, 4)(2) = (8, 0) \text{ m}$.



Answer. Equal speed means equal velocity magnitudes. It says nothing about whether the relative velocity points along the line that closes the relative-position vector.

Reasoning checkpoint. The compact collision test is whether one positive time makes $\vec{r}_{B/A}(0) + \vec{v}_{B/A}t = \vec{0}$.